

Advanced MIMO Technology

Presented By~

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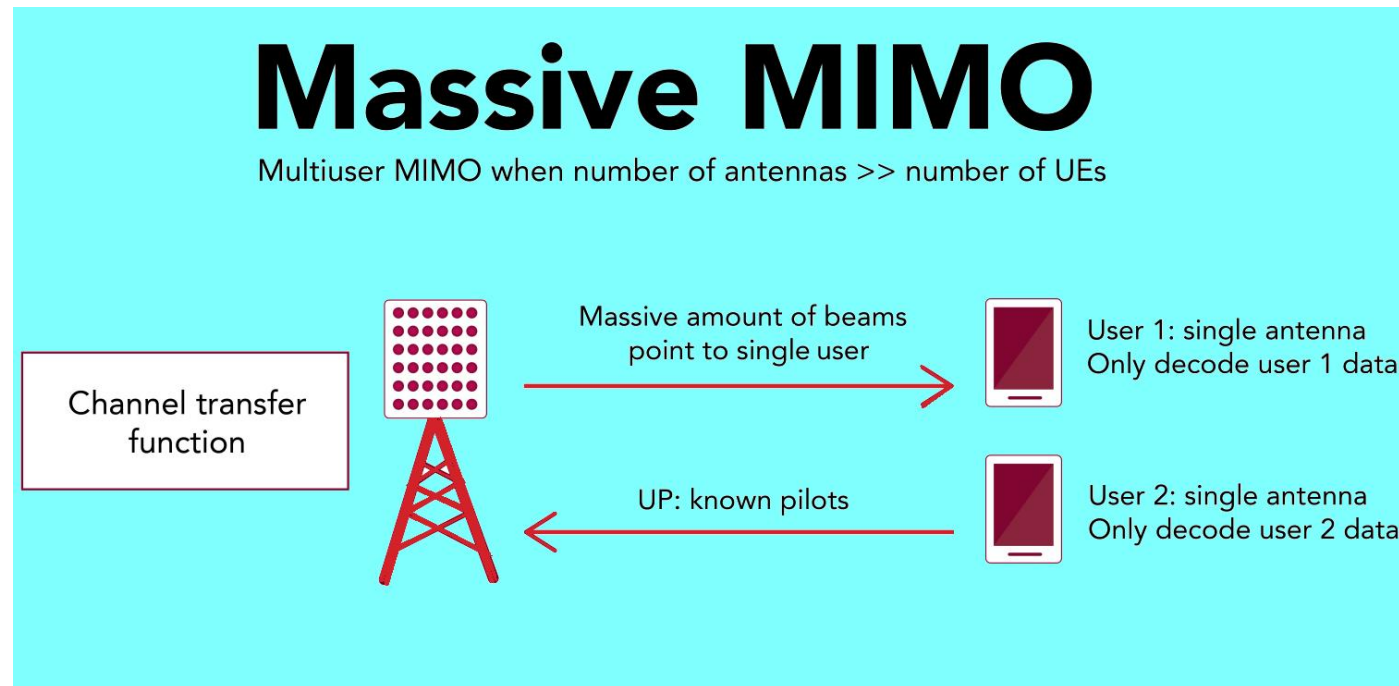
ICE, NSTU

Massive MIMO

- The MIMO technology is called a Multiple-Input Multiple-Output (MIMO) technology, and may also be called a Multiple Transmit Multiple Receive Antenna (MTMRA, Multiple Transmit Multiple Receive Antenna) technology.
- The basic principle is to use multiple transmitting antennas and receiving antennas at the transmitting end and the receiving end, respectively, and can distinguish signals sent to or from different spatial orientations and can also improve the system capacity without increasing the bandwidth and transmission power.
- Coverage and signal-to-noise ratio, improving the transmission quality of wireless signals, it differs from traditional signal processing methods in that it studies signal processing issues from both time and space.

Massive MIMO

- What are the working modes of MIMO?
- In today's environment where intelligent devices are increasing day by day, the application of MIMO technology is quite critical. We can adopt different working modes according to different conditions and different wireless environments. The agreement defines the following seven MIMO working modes:



Massive MIMO

- (1) **Single-antenna working mode:** It is also known as the SISO (Single-Input Single-Output) system, which uses a single antenna to transmit signals and a single antenna to receive signals.
- (2) **Open-loop transmit diversity:** Using the mathematical method of complex conjugates, orthogonal spatial channels are formed on multiple antennas, and the same data stream is sent to improve transmission reliability.
- (3) **Open-loop spatial multiplexing:** Artificially create "multipath effect" on different antennas, and one antenna transmits normally. Other antennas introduce a phase offset link. The transmission relationship of multiple antennas constitutes a complex matrix, and different data streams are transmitted in parallel. **This complex matrix is randomly selected at the transmitting end**, independent of the feedback result at the receiving end, which is open-loop spatial multiplexing.

Massive MIMO

(4) **Closed-loop spatial multiplexing:** When the transmitter transmits multiple data streams in parallel, it selects a complex matrix that creates "multipath effects" according to the feedback channel estimation results, which is closed-loop spatial multiplexing.

(5) **MU-MIMO:** Allows the transmitter to transmit data with multiple users at the same time.

(6) **Closed-loop RANK=1 precoding:** That is, space diversity technology. As a special case of closed-loop spatial multiplexing, only one data stream is transmitted, that is to say, the rank of the spatial channel is Rank=1. This working mode plays a role in improving transmission reliability, and is actually a way of transmitting diversity.

(7) **Beamforming:** Beamforming is also called a smart antenna. By phase-weighting the correlation of the output signals of multiple antennas, the signals form in-phase superposition in a certain direction and form phase cancellation in other directions, thereby achieving signal gain.

Massive MIMO

What are the advantages of MIMO technology?

(1) Increased channel capacity

The MIMO system can increase the channel capacity under the condition of high signal-to-noise ratio, and can be used in an open loop, that is, when the transmitting end cannot obtain channel information, and can also increase the information transmission rate without increasing the bandwidth and antenna transmission power. Thus, the spectrum utilization rate is greatly improved.

(2) Enhanced channel reliability

Using the space multiplexing technology provided by the MIMO channel can greatly enhance the stability of the system, and can also increase the transmission rate.

With the advent of the 5G era, the growth of long-distance communication traffic far exceeds the growth of wireless communication network capacity, and MIMO technology appropriately increases the number of antennas to increase channel capacity and reduce transmission power when the channel matrix is full. Satisfies the needs of wireless communication services. Therefore, the antenna has become an indispensable part in the subsequent evolution of wireless communication technology. Today's 5G technology also applies its subsequent evolution of Massive MIMO (massive MIMO technology), and has become a crucial part of 5G technology.

Massive MIMO

- ❖ Though MIMO technology relying on **large-scale phased arrays** is perceived to enhance the network capacity significantly by exploiting the **spatial diversity**, inherent limitations of **phased arrays** make them unable to fulfill such challenging visions of 6G networks.
 - ❖ By leveraging the holographic principle, the Reconfigurable Holographic Surface (RHS) can serve as a transmit/receive antenna integrated with the transceiver to generate beams with desirable directions.
 - ❖ The RHS utilizes the metamaterial to construct a holographic pattern on its surface based on the holographic interference principle.
- ❖ Antenna spacing in MIMO arrays is usually equal to half of the wavelength due to the implementation difficulties of manufacturing antennas.
 - Size of the radiation elements and the spacing between radiation elements of RHS (Holographic MIMO) can be much smaller than half of the wavelength, leading to superior beam-steering capabilities.
- Holographic MIMO (made of RHS) can be considered as a **potential solution** which is a transceiver controlled by software and generate **3D beams** following **holographic principle**.

Thus, we need new technology of RHS or others for 6G?

Massive MIMO

Beamforming:

- Beamforming is used with phased array antennae systems to focus the wireless signal in a chosen direction, normally towards a specific receiving device. This results in an improved signal at the user equipment (UE), and also less interference between the signals of individual UE. Phased antenna arrays are designed so that the radiation patterns from each individual element combine constructively, with those from neighboring elements forming an effective radiation pattern, the main lobe, which transmits energy in the desired direction.

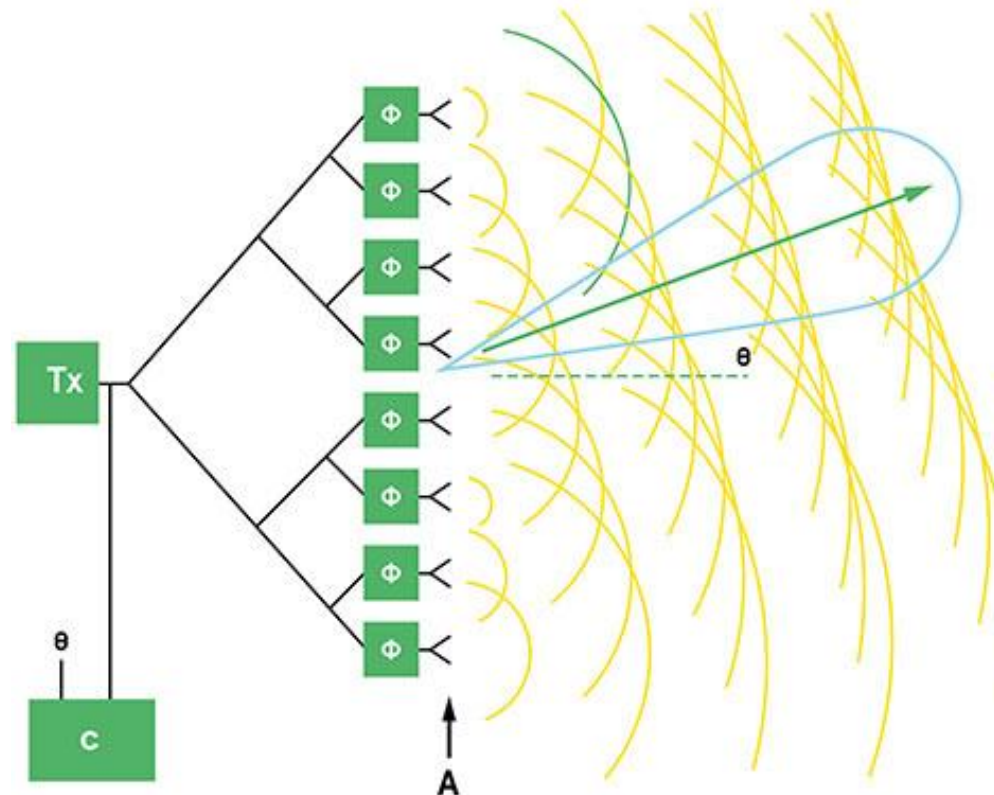
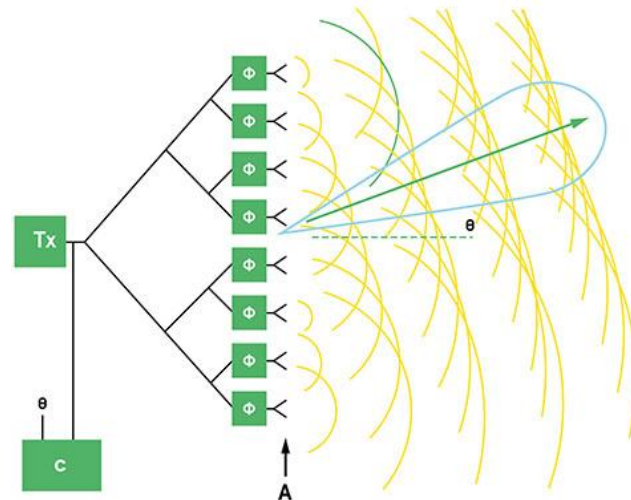


Fig. 1: Phased array antenna systems enable beamforming and steering

Massive MIMO

Beamforming:

- At the same time, the antenna array is designed so that signals sent in undesired directions destructively interfere with each other, forming nulls and side lobes. The overall antenna array system is designed to maximize the energy radiated in the main lobe, whilst limiting the energy in the side lobes to an acceptable level. The direction of the main lobe, or beam, is controlled by manipulating the radio signals applied to each of the individual antenna elements in the array.
- Each antenna is fed with the same transmitted signal but the phase and amplitude of the signal fed to each element is adjusted, steering the beam in the desired direction.
- Fast steering of the beam is achievable since the phase and amplitude of each signal are controlled electronically, allowing adjustments to be made in nanoseconds.



Different ways to improve system performance in Massive MIMO system:

- Integrating intelligent reflecting/refracting surface (IRS)
- Integrating reconfigurable holographic surface (RHS)
- Integrating intelligent omni surface (IOS)
- Flexible Intelligent Meta-surface (FIM)
- Stacked Intelligent Meta-surface (SIM)

Scenario I

Integration of IRS:

System Model:

- ❖ The proposed system considers a **MISO downlink communication system** that consists of a **BS equipped with a uniform linear array (ULA) of M antennas** and K single-antenna users.
- ❖ To better serve the users, a RIS constructed as UPA is employed with N reflection elements.
- ❖ To obtain the phase shifts configuration, the RIS is **linked to an intelligent controller** which communicates with the BS via a separate link.

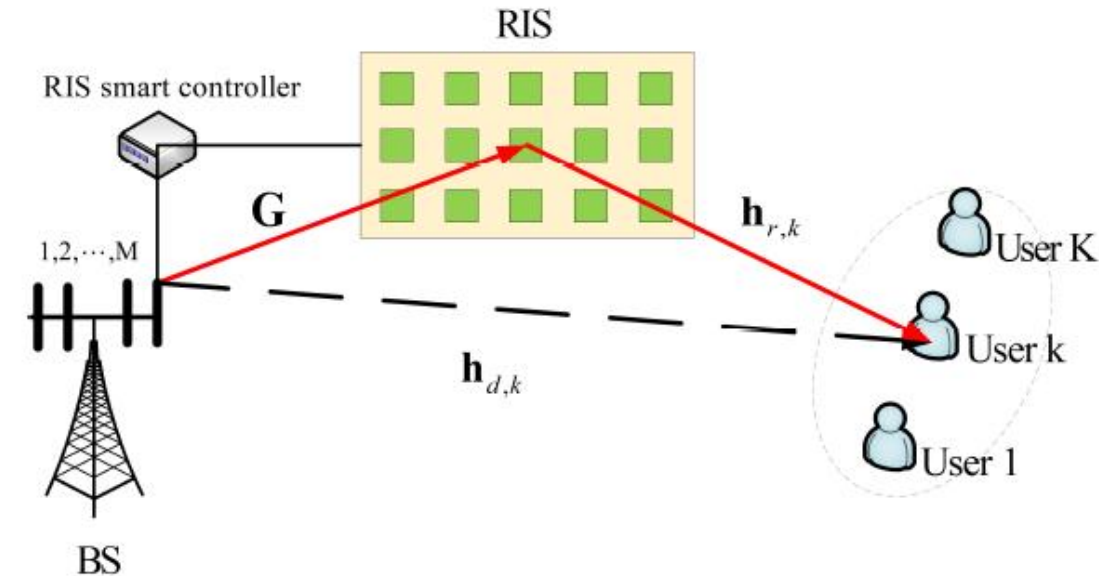


Fig. 1. RIS-assisted multi-user MISO communication system.

Scenario II Integration of RHS:

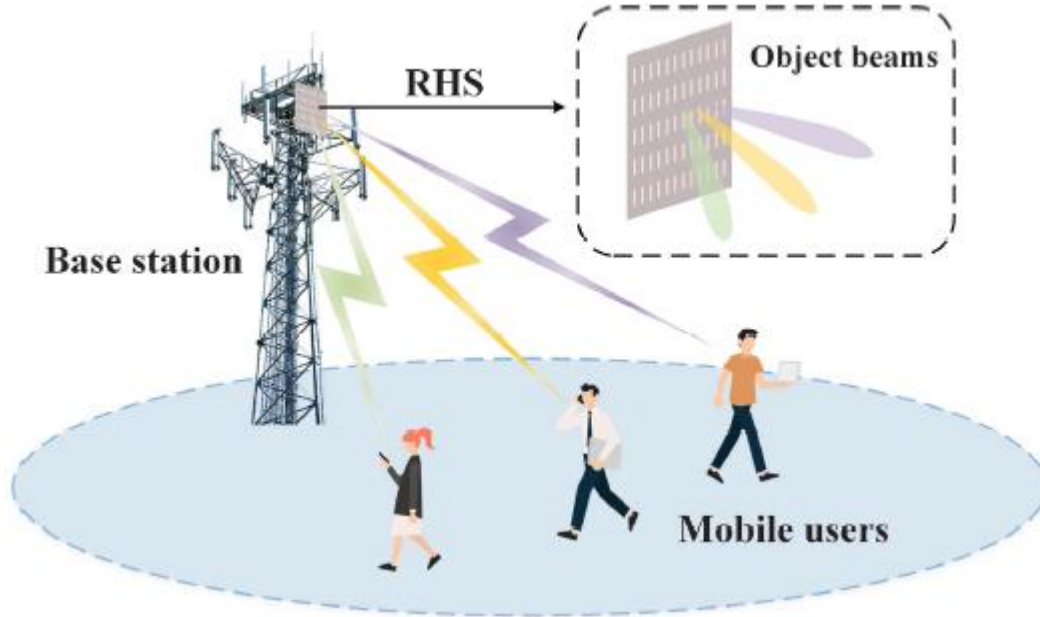


Fig. 1. Reconfigurable holographic surface-aided multi-user communication system.

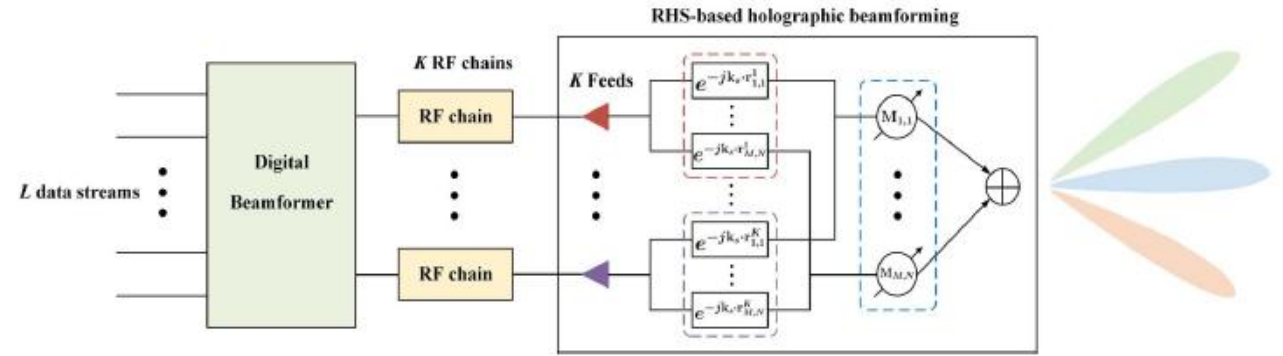
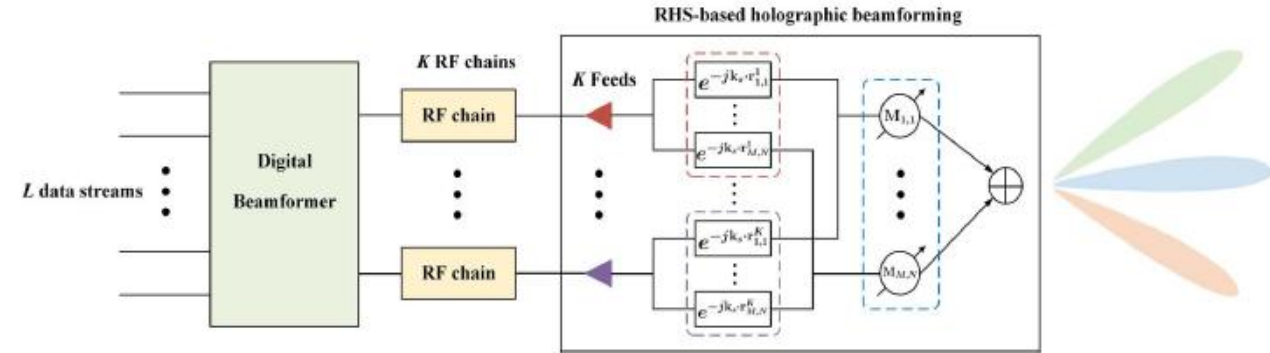
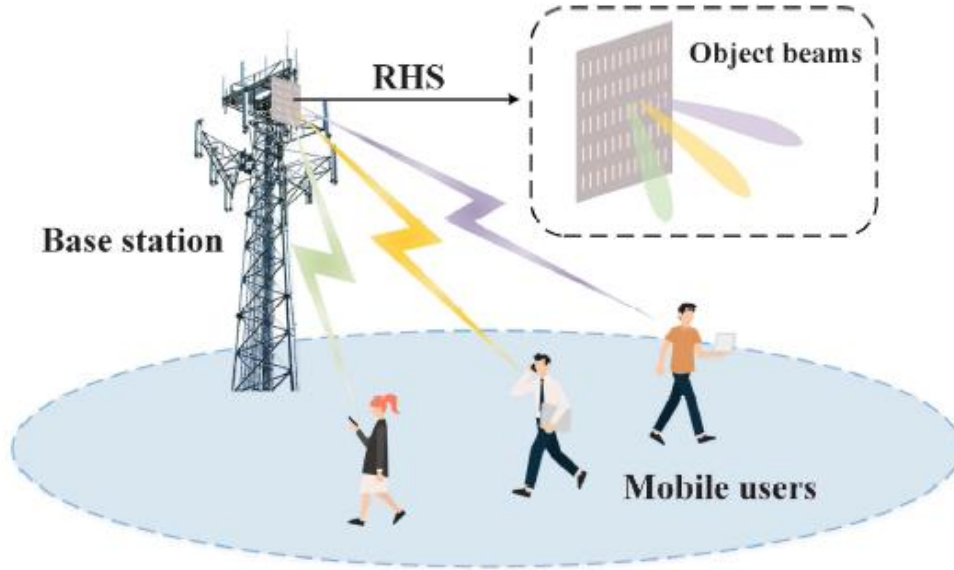


Fig. 2. Block diagram of the hybrid beamforming.

- ❖ The BS encodes L different data streams via a digital beamformer $\mathbf{V} \in \mathbb{C}^{K \times L}$, and then up-converts the processed signals to the carrier frequency by passing through RF chains.
- ❖ Each RF chain sends the up-converted signals to its connected feed and the feed then transforms the high frequency current into an electromagnetic wave (reference wave) propagating on the RHS.
- ❖ The radiation amplitude of the reference wave at each element is controlled via a holographic beamformer $\{\mathbf{M}_{m,n}\}$ to generate desired beams.

Scenario II

Integration of RHS:



- ❖ To achieve accurate multi-beam steering, it works as a hybrid beamforming scheme where the BS performs digital beamforming, the RHS performs holographic beamforming by leveraging the holographic technique, and each user conducts receive combining.

Scenario III

Integration of RHIS

System Model:

- ❖ Sensing and communication functions are executed in three steps: Optimization, Transmission, and Reception.
- Optimization: The BS optimizes the ISAC signals to maximize the radar performance and to assure the target quality of service (QoS) to communication users.
- Transmission: The RHS transmits the ISAC signals with multiple beams towards the directions of the users and targets.
- Reception: The communication users receive the signals from the RHS and decode the received signals to obtain the communication information. At the same time, the MIMO antenna array listens to the echo signals reflected by the targets for radar sensing.

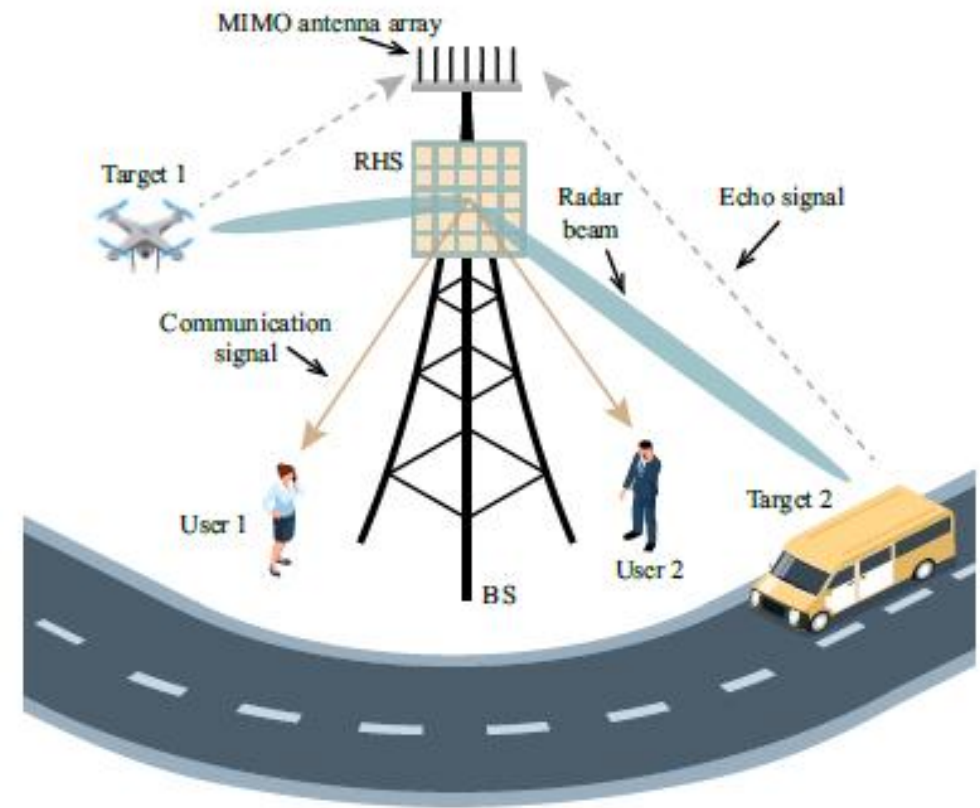


Fig. 1. A holographic integrated sensing and communication system.

Scenario IV

Holographic MIMO:

System Model:

- ❖ The proposed system includes **three steps**:
 - **Sensing of information** by holographic grid array (HGA) utilizing **communication and echo signals**.
 - Transmission of **desired beamforming** for the heterogeneous users.
 - Reception of **signals and services** by the users.
- The radiating elements of holographic grid array (HGA) are capable to **control the radiation amplitude of the reference signals** with a view to producing **desired beamforming** for serving the users.
- **Minimum number of grids/radiating elements** from HGA can be activated for beamforming based on user requirements.
- Ensuring the effective beamforming **maximizing utility function for sensing** considering beampattern gain, user distances, and **sensing-communication losses**.
- Maximizing **SINR and minimizing power consumption** for serving users.

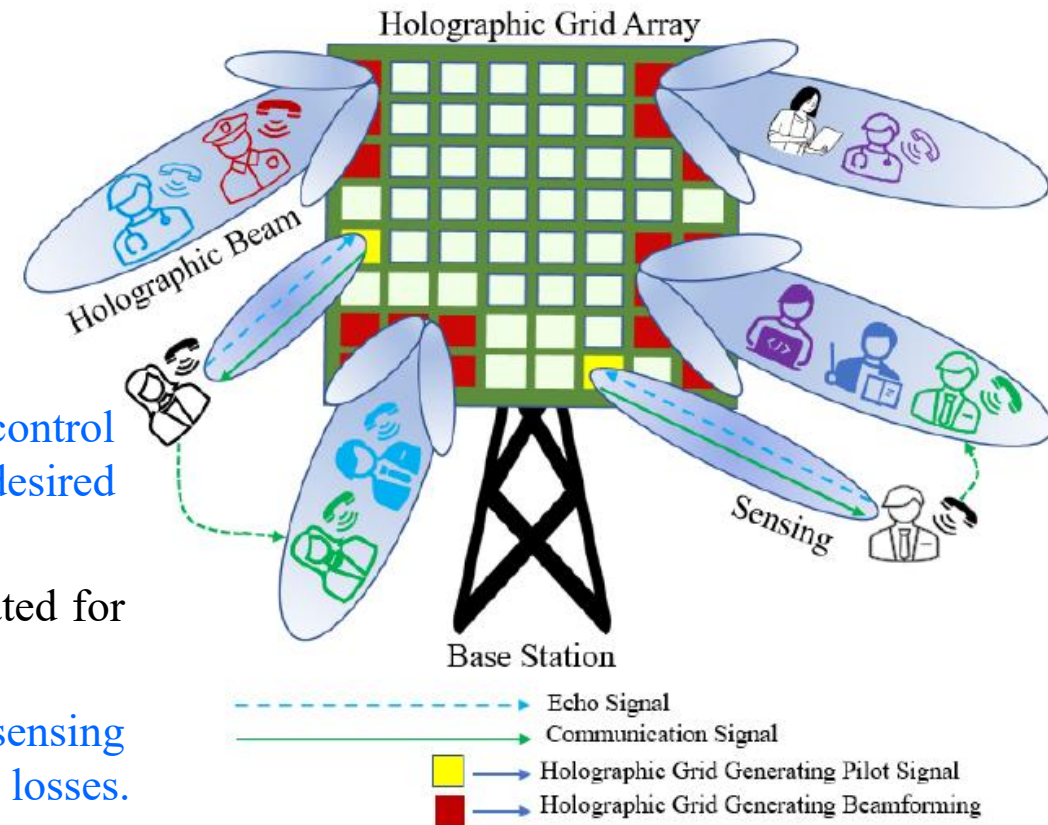


Fig. 1. A System model for target oriented integrated sensing and communication in holographic MIMO.

Scenario V

Holographic MIMO (HMIMO) and Cell Free Network:

System Model:

- ❖ The proposed HMIMO-aided CF network (HACN) system decreases the necessity for a large number of grids due to the integration of the lower dimensional HMIMOs replacing BSs in the CF network, as demonstrated in Fig. 1.
- ❖ The proposed HACN system includes three steps:
 - Step 1: The serving HMIMOs having inactivated grids activate the required number of grids from the respective HGAs to allocate the required power for beamforming to serve the corresponding users in the system.
 - Step 2: The power is added that is allocated from the serving HMIMOs for beamforming to serve the users.
 - Step 3: The users get the desired services with the required power allocation for beamforming.

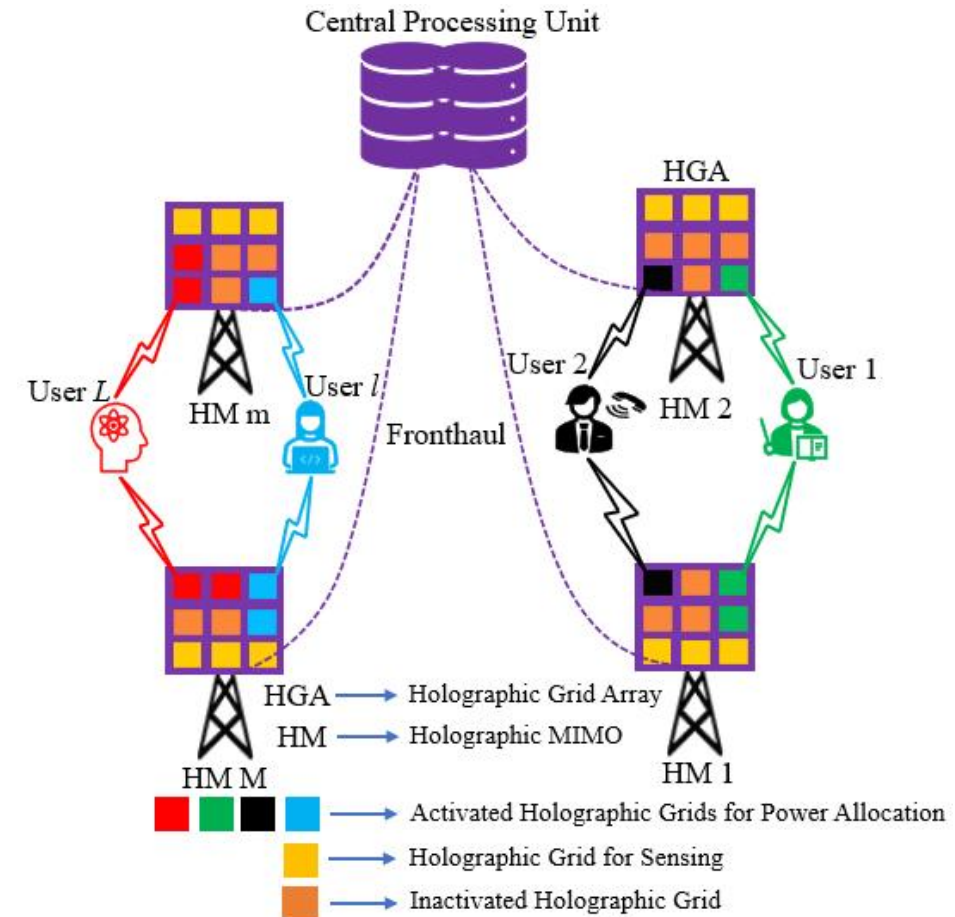


Fig. 1. A system model for power allocation in holographic MIMO-aided cell-free network.

Scenario V

Holographic MIMO (HMIMO) and Cell Free Network:

- Cell-free massive MIMOs (CFMMs) and holographic MIMOs (HMIMOs) are potential contenders to enable an energy-efficient intelligent networking system that ensures power savings and provides the necessary beamforming for serving the users.
- HMIMO can be considered as a potential solution which is a software-controlled transceiver that uses the holographic principle to create 3D beams having a holographic grid array (HGA) installed on the top of base stations (BSs).
- A CFMM network is made up of a large number of self-controlled BSs or access points (APs) that are distributed over a large geographical area for providing services to multiple users simultaneously.
- Therefore, the integration of HMIMOs replacing APs in CF networks guarantees an energy-efficient CF network that ensures power savings for serving users.
- The required number of serving HMIMOs allocate the desired power for serving the users by activating the required number of grids from the respective HGAs ensuring savings of power and improvement of energy efficiency.

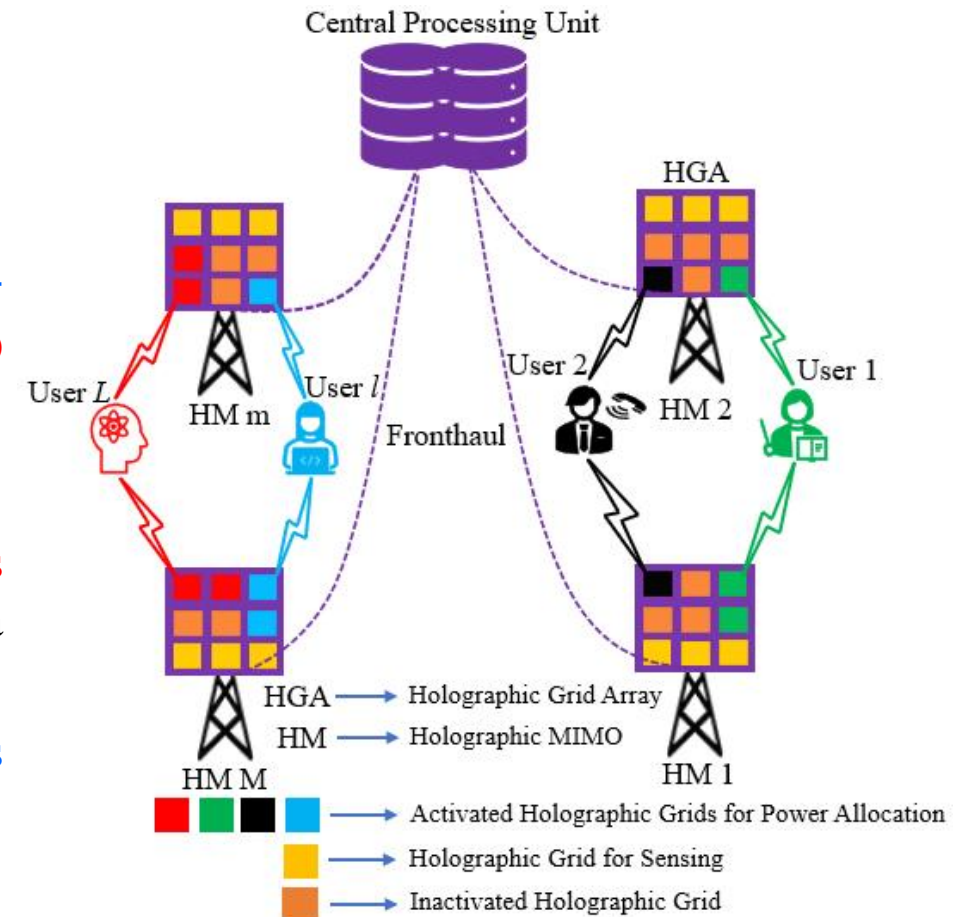


Fig. 1. A system model for power allocation in holographic MIMO-aided cell-free network.

Scenario VI

Integration of IOS:

Network Description:

- ❖ The IOS is integrated in the Cell-Free Massive MIMO (CFMM) system which can guide the reflected ray and refracted ray towards the users who are positioned on both the sides of the IOS.
- ❖ Consequently, we can achieve the same capacity with half number of IOS to serve users.
- ❖ Each BS only serves a certain number of users, and some of the BSs give services to each user.
- ❖ The IOSs optimize the required phase shift either using reflective channel or refractive channel or both which improves the quality of the services.

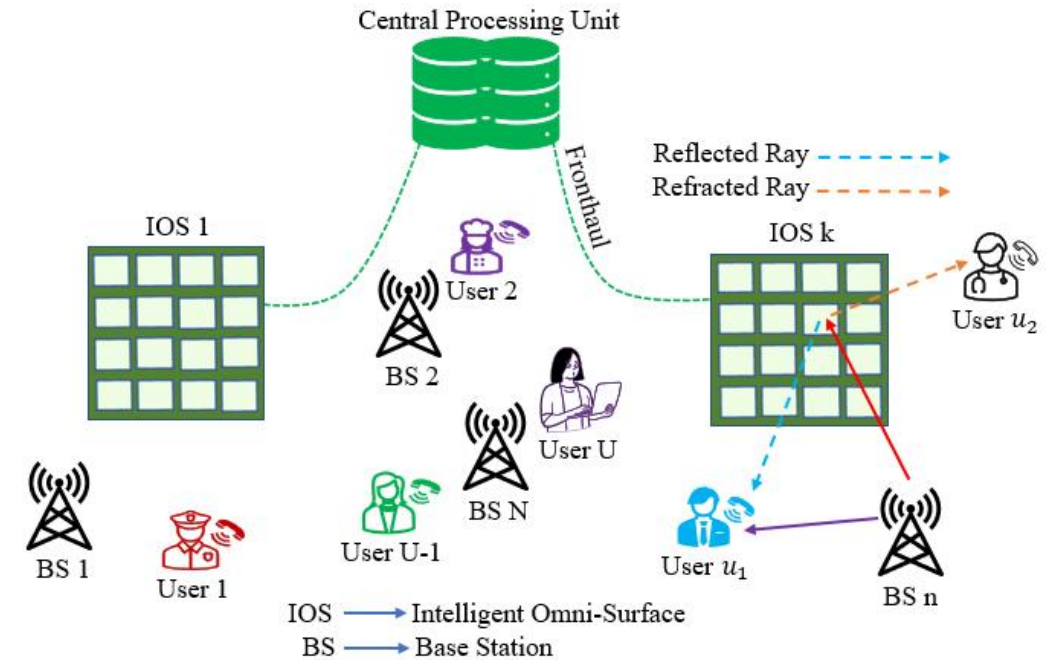


Fig. 1. A system model for intelligent omni surface-assisted cell-free massive MIMO system.

Scenario VI

Integration of IOS:

❖ Characteristics of IRS/IOS:

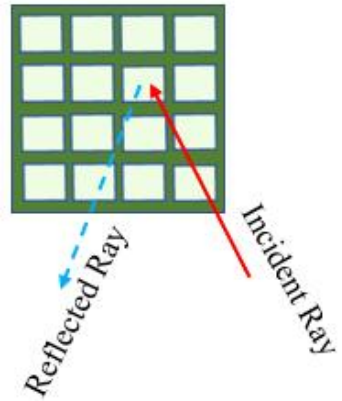


Fig 1(a). Intelligent Reflecting Surface

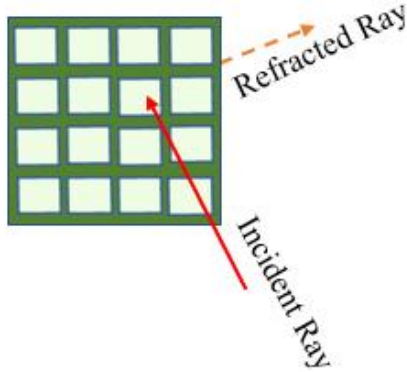


Fig 1(b). Intelligent Refracting Surface

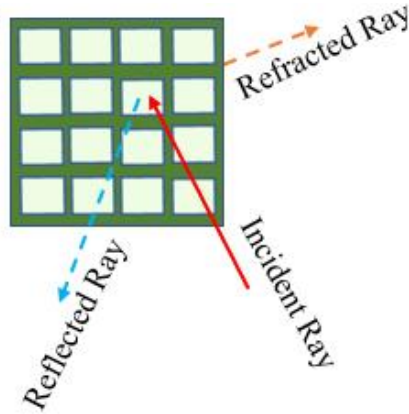
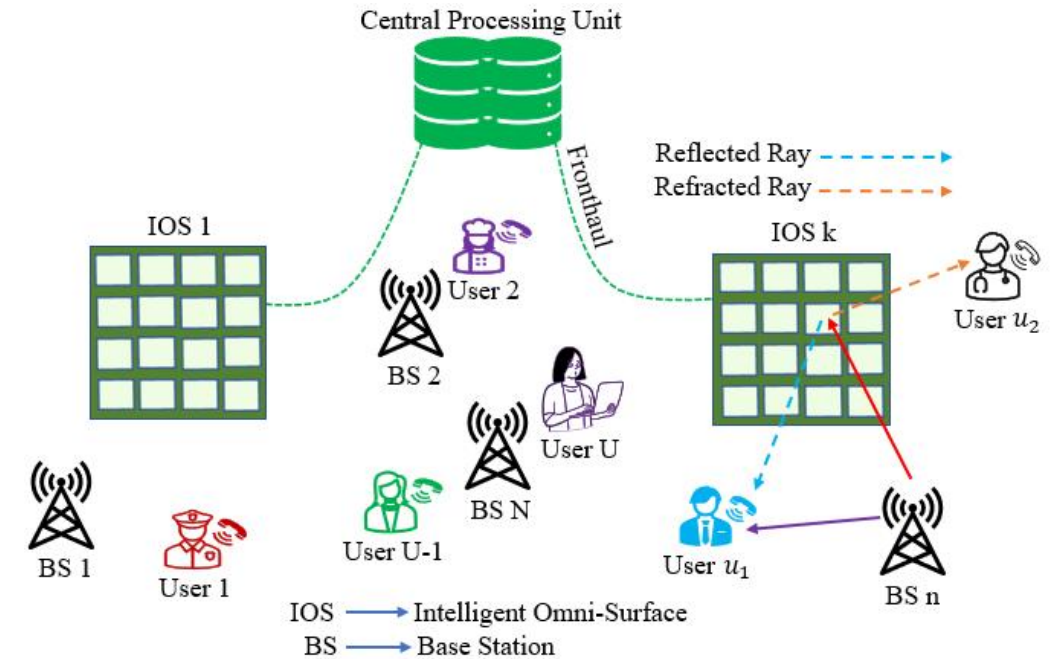


Fig 1(c). Intelligent Omni-Surface



Flexible Intelligent Meta-surface (FIM):

- A flexible intelligent metasurface (FIM) is a reconfigurable antenna array made of low-cost elements that can both adjust their electromagnetic (EM) response and change their 3D shape to optimize wireless signals.
- Unlike rigid surfaces, FIMs can dynamically morph their form to adapt to the environment, which improves signal quality, energy efficiency, and spectral efficiency in applications like wireless sensing and communication.

Stacked Intelligent Meta-surface (SIM):

- A Stacked Intelligent Metasurface (SIM) is a multi-layered structure that processes electromagnetic (EM) waves in real-time.
- By stacking multiple programmable metasurface layers, SIMs can perform complex signal processing tasks like beamforming, MIMO precoding, and interference mitigation directly on the EM wave itself, eliminating the need for bulky and energy-intensive RF chains.
- This technology is expected to be key for future wireless communication systems, offering enhanced performance and efficiency.

Thank You
Best Wishes for Exam!